Supplementary material for UAI 2026 Submission 642
Selected results — BASE hierarchy and BaVarIA (32 reps)

This summary surfaces a small selection of figures and tables from the companion reports `base32.pdf` and `bavaria32.pdf`. Each panel shows two representative testbeds side-by-side: CIFAR-100 / WRN and Texas / MLP3 throughout. All numbers are means over 32 leave-one-out replicates; bands and error bars are ±1 SE.

1 BASE hierarchy

Headline. The BASE1→BASE4 ordering reverses with K — simpler families dominate at small K, richer families at large K (quantified in concordance Table 1 below).

Technical note. The submitted BASE2 takes pooled mean and variance to score each sample as

$$\text{BASE2} = \frac{z_i - \hat{\mu}_{\text{pool}}}{\hat{\sigma}^2},$$

which showed poor performance accross all shadow model budgets. Here, we additionally evaluate an alternative two parameter attack we denote BASE2*,

$$\text{BASE2}^* \propto (\hat{\mu}_{i,1} - \hat{\mu}_{i,0})\big(z_i - \frac{1}{2}(\hat{\mu}_{i,1} + \hat{\mu}_{i,0})\big),$$

which keeps separate IN/OUT means but a single constant variance shared across samples.

The introduction of BASE2* as an intermediate point was prompted by a reviewer's pointer to IMIA (Du et al., 2025), whose non-parametric scoring variant is equivalent to this form; we propose folding BASE2* into the hierarchy in the camera-ready. It is derived from the Gaussian model as a parameter constrained and simplified alternative that fits naturally between BASE3 and BASE1.

Observations.

- The hierarchy is upheld at $K = 8$ (simple wins) and reversed at $K = 254$ (complex wins). The pattern holds for $\geq 10/12$ testbeds in every tail metric.
- BASE4 (the most parameter-rich variant; equivalent to LiRA online) collapses at $K \leq 8$ on TPR@FPR= 0.01, exactly the regime where the simpler BASE1 / BASE2* dominate — consistent with per-class variance estimation being the bottleneck.
- BASE2* tracks BASE3 closely on AUC at every K, so the per-sample variance estimate adds little to AUC; the gap between them appears mainly at lower FPR, where BASE3's per-sample variance helps in the deep tail.
- The submitted BASE2 (pooled mean and variance) sits near the floor across all K on both panels, so the submitted hierarchy lacks a competitive two-parameter intermediate. BASE2* is introduced to fill this gap.

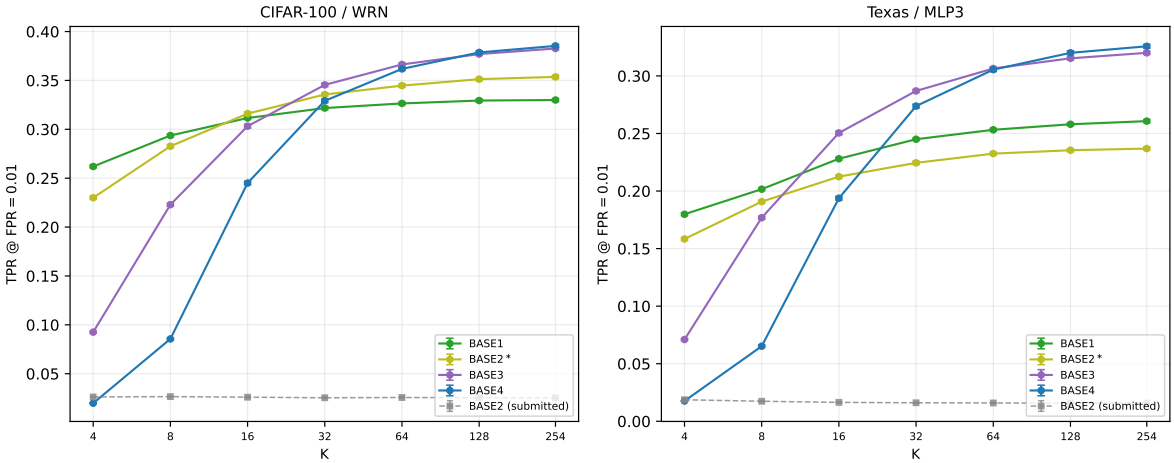

Figure 1: TPR @ FPR= 0.01 as a function of K. BASE1, BASE2*, BASE3, BASE4 (the four headline curves) plus the submitted BASE2 (dashed grey, low-prominence). BASE2 sits near the floor at TPR@0.01 ≈ 0.02 across all K on both panels, illustrating why we add BASE2* as a more informative intermediate point. Selected from `base32.tex` Fig. 3 (panels b and f).

	$K = 8$ *(simple wins)*			$K = 254$ *(complex wins)*		
Testbed	AUC	TPR@.01	TPR@.001	AUC	TPR@.01	TPR@.001
CIFAR-10 / WRN	**+1.00**	**+1.00**	**+1.00**	**+1.00**	**+1.00**	**+1.00**
CIFAR-100 / WRN	+0.71	**+1.00**	**+1.00**	**+1.00**	**+1.00**	**+1.00**
CINIC-10 / WRN	+0.08	+0.74	+0.79	**+1.00**	**+1.00**	**+0.94**
Location / MLP3	*-0.53*	+0.75	**+0.99**	**+1.00**	**+1.00**	**+1.00**
Purchase / MLP3	**+1.00**	**+1.00**	**+1.00**	+0.54	+0.83	+0.60
Texas / MLP3	**+1.00**	**+1.00**	+0.66	**+1.00**	+0.85	+0.73
CIFAR-10 / RN18	**+1.00**	**+1.00**	**+1.00**	*-0.54*	+0.71	+0.86
CIFAR-100 / RN18	-0.41	+0.38	+0.82	**+1.00**	**+1.00**	**+1.00**
CINIC-10 / RN18	-0.35	+0.55	+0.90	**+1.00**	**+1.00**	**+1.00**
Location / MLP4	-0.45	+0.37	+0.89	**+1.00**	**+0.99**	**+0.94**
Purchase / MLP4	**+1.00**	**+0.93**	+0.79	+0.90	**+0.90**	**+0.92**
Texas / MLP4	+0.57	**+1.00**	**+1.00**	**+1.00**	**+0.97**	+0.76
Mean	+0.39	+0.81	**+0.90**	+0.82	**+0.94**	+0.89
Median	+0.64	**+0.97**	**+0.94**	**+1.00**	**+1.00**	**+0.94**

Table 1: Weighted concordance $S \in [-1, +1]$ of the observed method ordering with the predicted BASE hierarchy. Reference at $K = 8$ is BASE1 > BASE2* > BASE3 > BASE4 (simple wins); at $K = 254$ it is the reverse (complex wins). $S = +1$ means all pairs agree weighted by AUC gap, 0 random, -1 fully reversed. **Bold**: $S \geq 0.9$. *Italic*: $S \leq -0.5$.

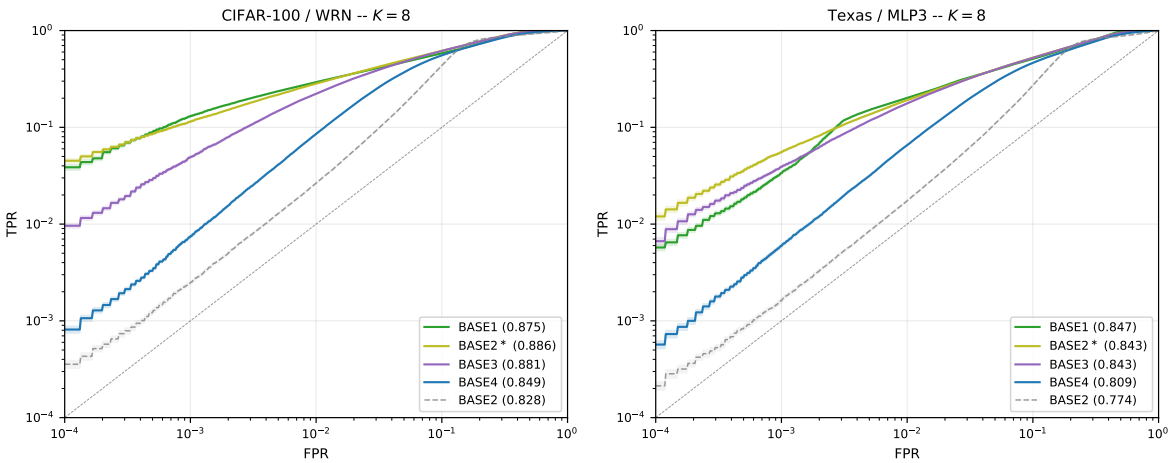

Figure 2: Online ROC at $K = 8$. BASE1, BASE2*, BASE3, BASE4 plus the submitted BASE2 (dashed grey, low-prominence). Selected from `base32.tex` Fig. 7 (panels b and f).

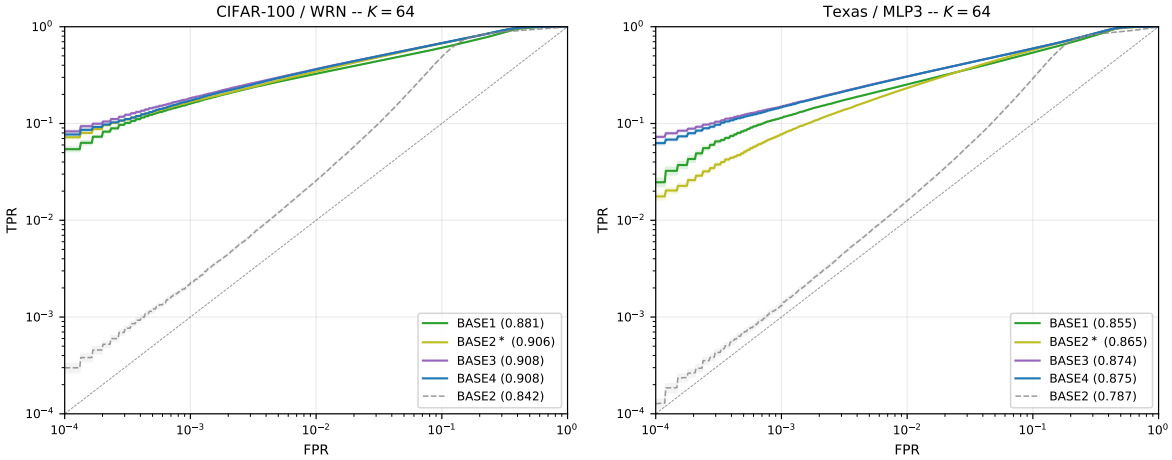

Figure 3: Online ROC at $K = 64$. Selected from `base32.tex` Fig. 9 (panels b and f).

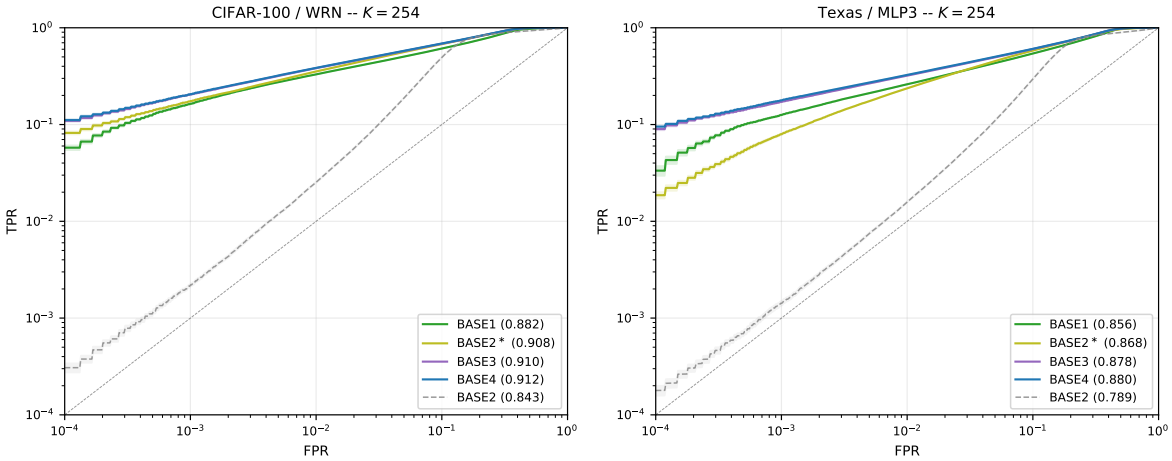

Figure 4: Online ROC at $K = 254$. Selected from `base32.tex` Fig. 11 (panels b and f).

2 LiRA vs. BaVarIA

BaVarIA is a plug-in for LiRA: it replaces LiRA's pooled-vs-per-sample variance rule with a single NIG-shrinkage estimator (continuous in K), and uses the NIG prior in place of the IN-class likelihood when offline shadow counts are zero. At $K = 64$ (around LiRA's variance-rule cutover) BaVarIA-n strictly dominates LiRA in TPR@FPR$= 0.01$ on 12/12 testbeds online and 10/12 offline; the offline gap persists at every K because LiRA's offline heuristic discards the IN-class likelihood while BaVarIA's offline mode substitutes the NIG prior. Per-testbed numbers in `bavaria32.pdf`.

Online

Headline. BaVarIA-n's smooth variance interpolation gives a small, persistent edge over LiRA across K, peaking around the K=64 rule cutover (Fig. 6 and table below).

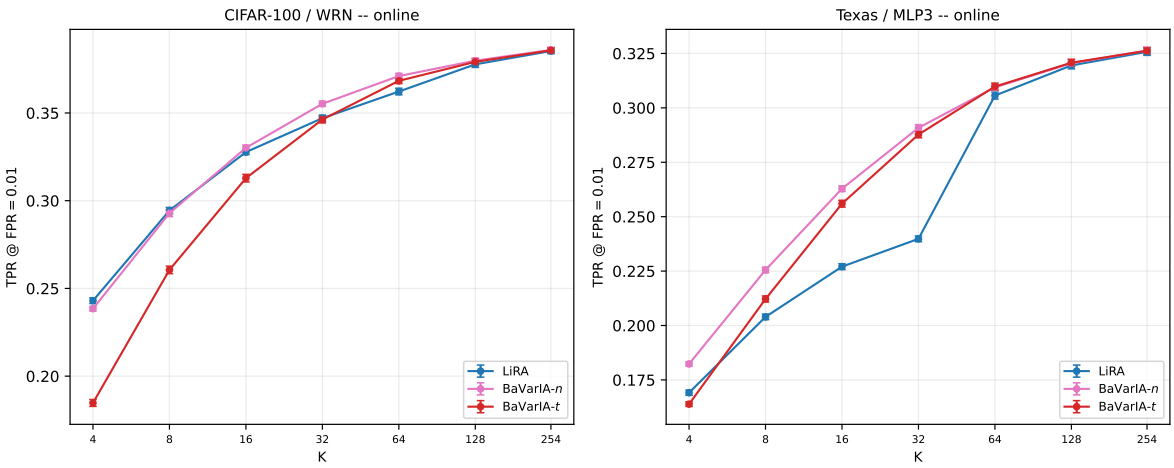

Figure 5: Online TPR @ FPR$= 0.01$ vs. K. LiRA, BaVarIA-n, BaVarIA-t. Selected from `bavaria32.tex` Fig. 3 (panels b and f).

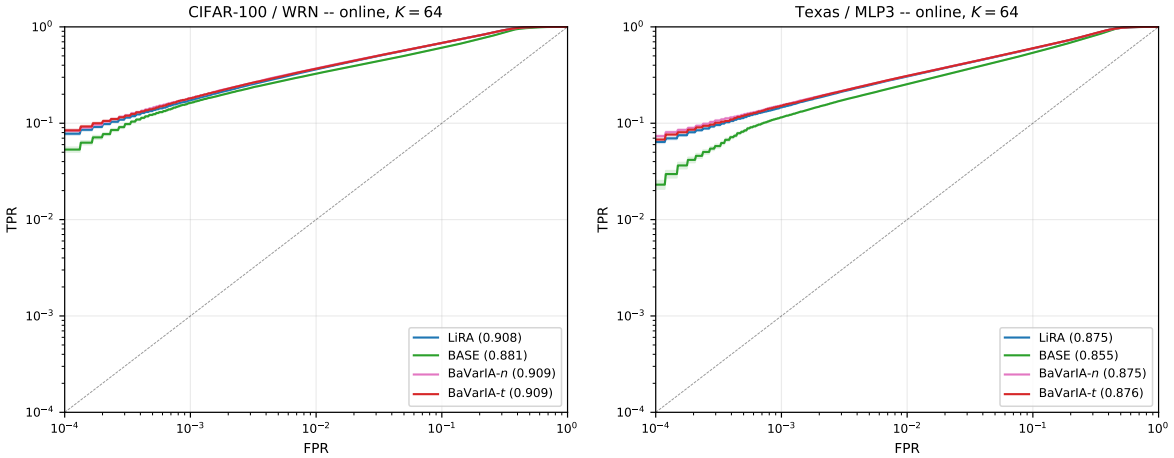

Figure 6: Online ROC at $K = 64$. LiRA, BASE, BaVarIA-n, BaVarIA-t. Selected from `bavaria32.tex` Fig. 7 (panels b and f).

Offline

Headline. BaVarIA's NIG-prior IN-class likelihood gives a persistent gap over LiRA (Fig. 7, 8).

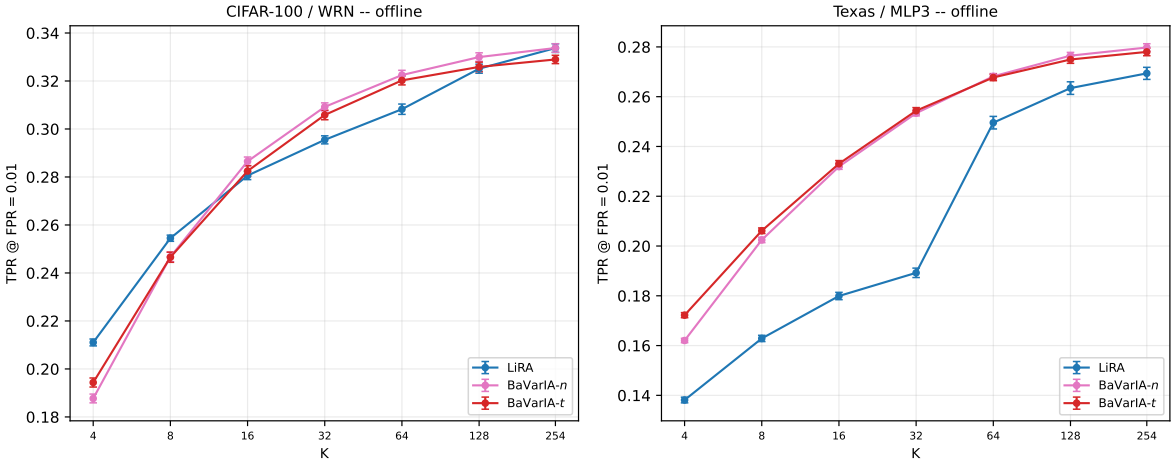

Figure 7: Offline TPR @ FPR= 0.01 vs. K. Selected from `bavaria32.tex` Fig. 13 (panels b and f).

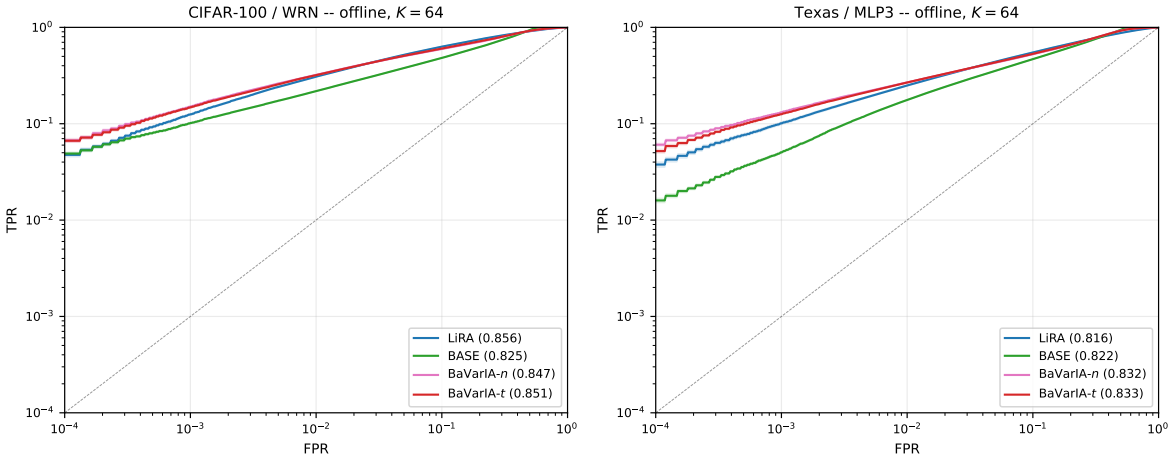

Figure 8: Offline ROC at $K = 64$. Selected from `bavaria32.tex` Fig. 17 (panels b and f).

Observations

- At $K = 64$, BaVarIA-n strictly dominates LiRA in TPR@FPR= 0.01 on 12/12 testbeds online and 10/12 offline; the offline gap persists at every K. By $K = 254$ online the three Gaussian methods converge by construction, so BaVarIA's value sits in the intermediate-K regime.
- The two BaVarIA variants split work along the metric: BaVarIA-t (heavier-tailed posterior predictive) tends to lead on AUC at small K, where per-sample means are noisy; BaVarIA-n tends to lead on TPR@FPR= 0.01.
- Offline, the gap over LiRA is largest at small K (where LiRA's offline heuristic is least informative) and shrinks but does not close as K grows, because LiRA-offline never sees the IN class while BaVarIA substitutes the NIG prior for it.

Testbed	LiRA	BaVarIA-n	BaVarIA-t
CIFAR-10 / WRN	0.129	**0.134**	0.133
CIFAR-100 / WRN	0.362	**0.371**	0.368
CINIC-10 / WRN	0.180	**0.187**	0.187
Location / MLP3	0.423	**0.440**	0.439
Purchase / MLP3	0.031	0.035	**0.036**
Texas / MLP3	0.306	0.309	**0.310**
CIFAR-10 / RN18	0.039	**0.042**	**0.042**
CIFAR-100 / RN18	0.663	0.669	**0.669**
CINIC-10 / RN18	0.288	**0.293**	0.291
Location / MLP4	0.352	**0.364**	0.361
Purchase / MLP4	0.053	0.059	**0.059**
Texas / MLP4	0.358	**0.359**	0.359

Table 2: Online TPR @ FPR= 0.01, $K = 64$. Mean over 32 replicates. Best per row in bold.

Testbed	LiRA	BaVarIA-n	BaVarIA-t
CIFAR-10 / WRN	**0.087**	0.074	0.069
CIFAR-100 / WRN	0.308	**0.322**	0.320
CINIC-10 / WRN	0.123	**0.181**	0.178
Location / MLP3	0.327	**0.412**	0.405
Purchase / MLP3	0.023	**0.026**	0.023
Texas / MLP3	0.250	**0.268**	0.268
CIFAR-10 / RN18	**0.038**	0.032	0.031
CIFAR-100 / RN18	0.579	0.663	**0.664**
CINIC-10 / RN18	0.223	**0.277**	0.273
Location / MLP4	0.299	**0.333**	0.326
Purchase / MLP4	0.038	**0.042**	0.039
Texas / MLP4	0.294	**0.321**	0.316

Table 3: Offline TPR @ FPR= 0.01, $K = 64$. Mean over 32 replicates. Best per row in bold.